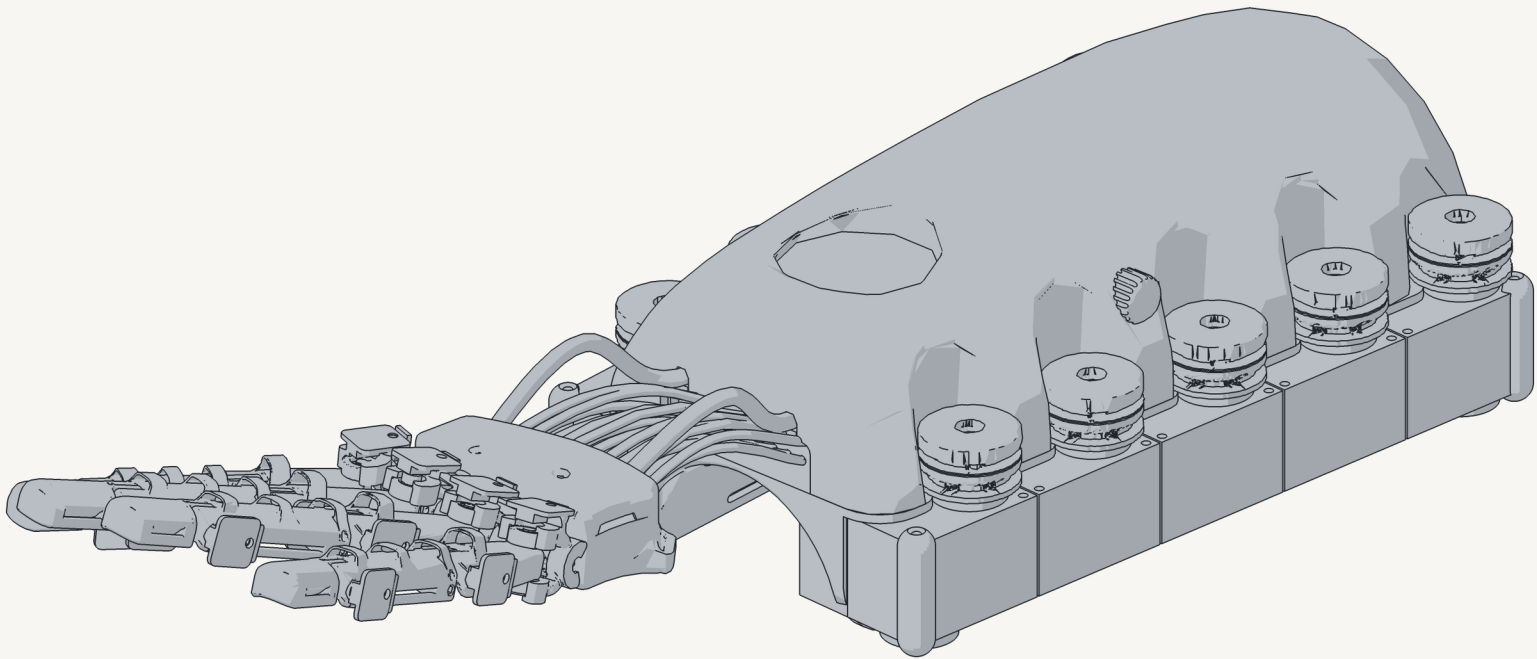


TAKTO ONE

A four-finger instrumented hand exoskeleton,
built and documented for assembly by hand.



HOW TO READ THIS

 the part you add now

 already built

Straight arrow push it that way

Curved arrow turn it that way

TURN + SLIDE touch first, then turn

MOVES FREELY? test before going on

DIAMETRIC ONLY magnet orientation matters

AS BUILT: 2 OF 10 budget, not design

Build one finger completely and check it moves before starting the other three. The finger chapter is drawn for the index; the other three are identical in sequence and differ only in link length.

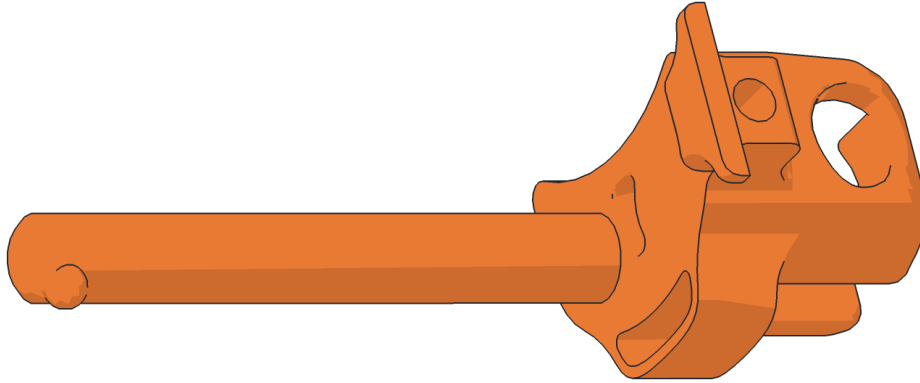
BUILD ONE FINGER

Drawn for the index finger. The other three are the same sequence with different link variants; build one completely and check it moves before the next.

1

START HERE

Start with the MCP link, thick half

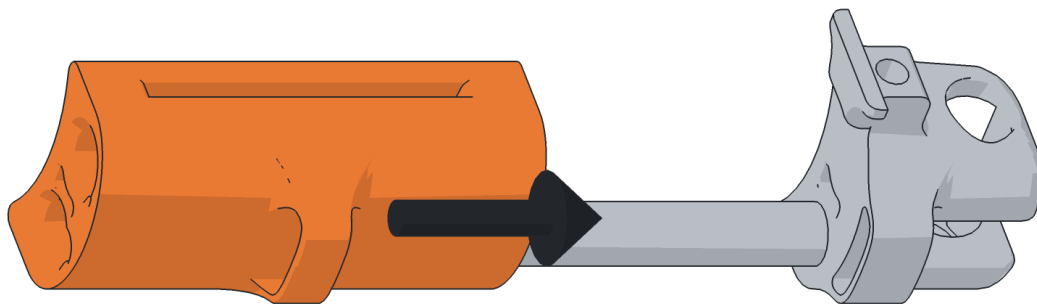


2

CLICKS PAST

SPRINGS 2.0 MM³

Fit the MCP centre piece

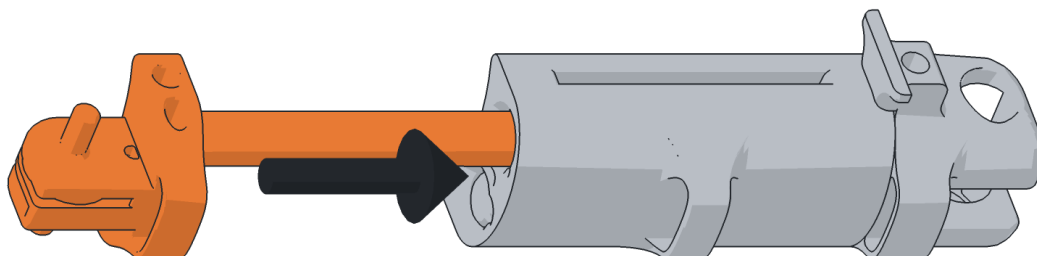


3

CLICKS PAST

SPRINGS 2.38 MM³

Close with the MCP thin half



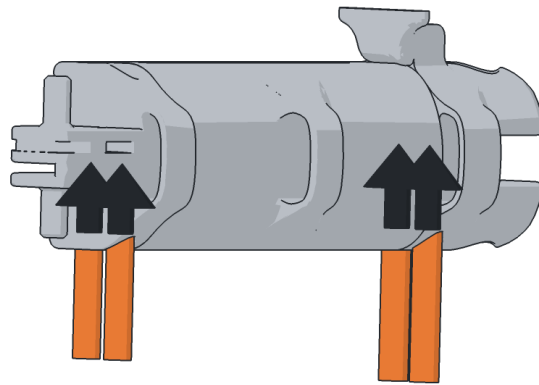
4

CLICKS PAST

4X Ø/2 STEEL, FLUSH

SPRINGS 1.62 MM³

Slide the guide rods flush

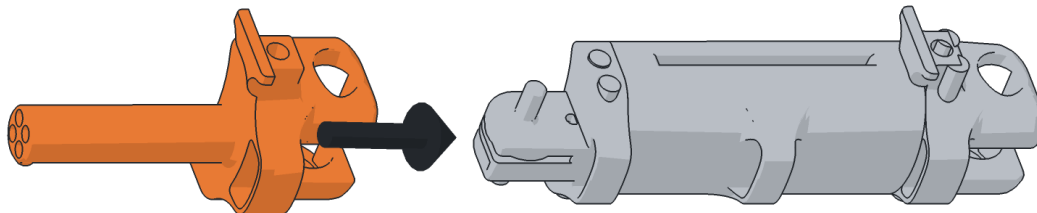


5

CLICKS PAST

SPRINGS 2.0 MM³

Slide the knuckle base on from behind

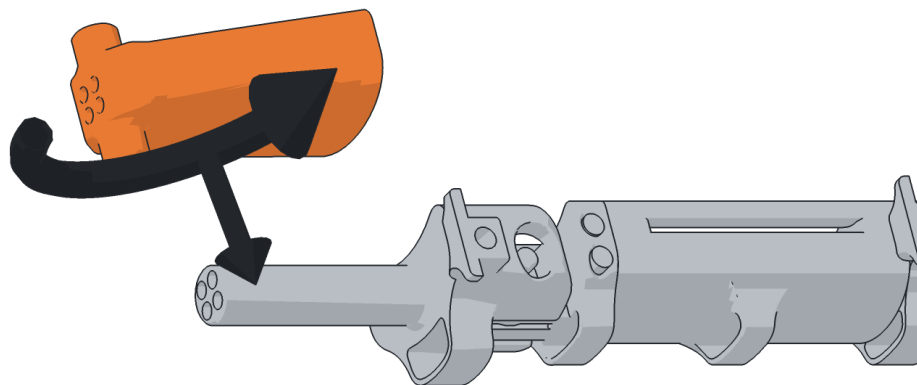


6

TURN AS YOU PUSH

SPRINGS 2.62 MM³

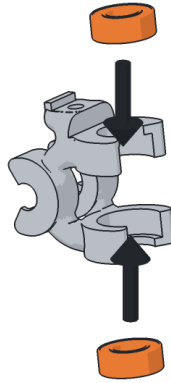
The knuckle plate from below, turning



7

2X MR84 4 X 8 X 3

Press both bearings into the root mount

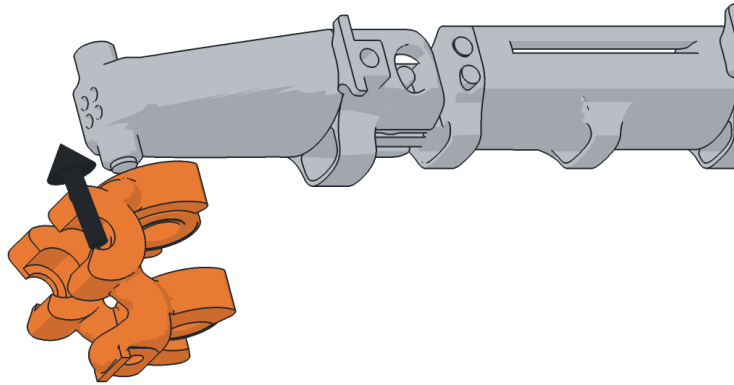


8

CLICKS PAST

SPRINGS 9.12 MM³

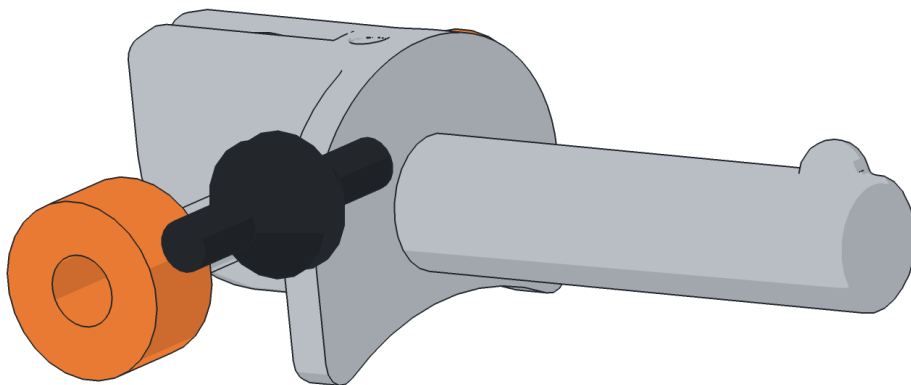
Lower the root mount over the shaft



9

2X 2 X 5 X 2.5 MM

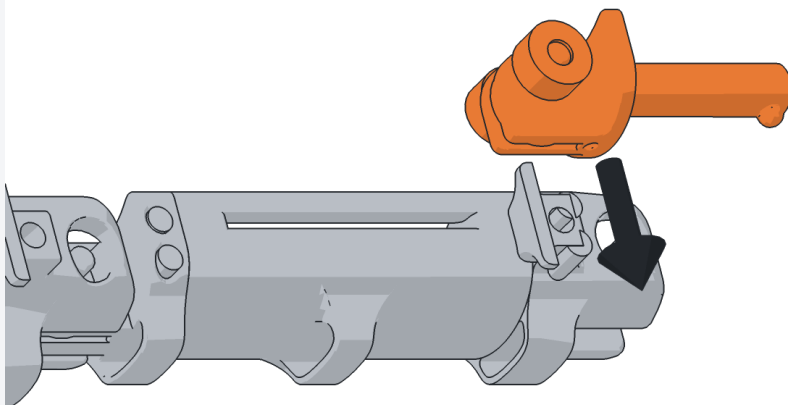
Press the bearings onto the PIP link



10

CLICKS PAST SPRINGS 2.0 MM³

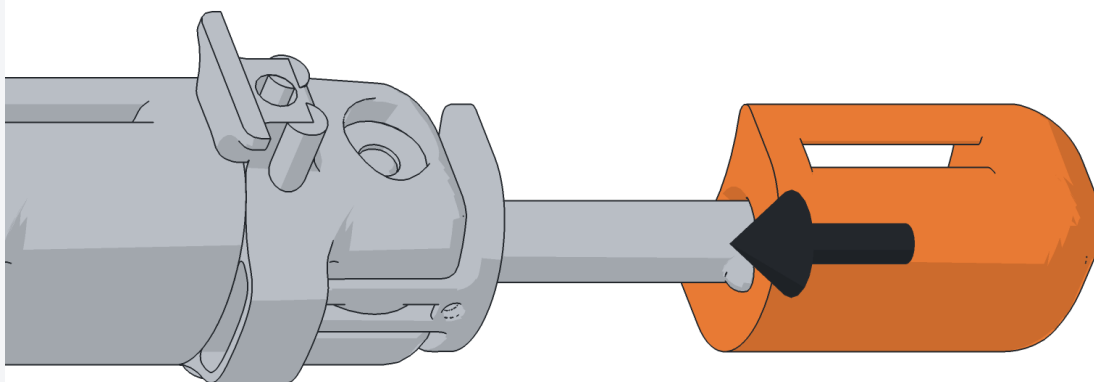
Fit the PIP link, bearings on



11

CLICKS PAST SPRINGS 2.62 MM³

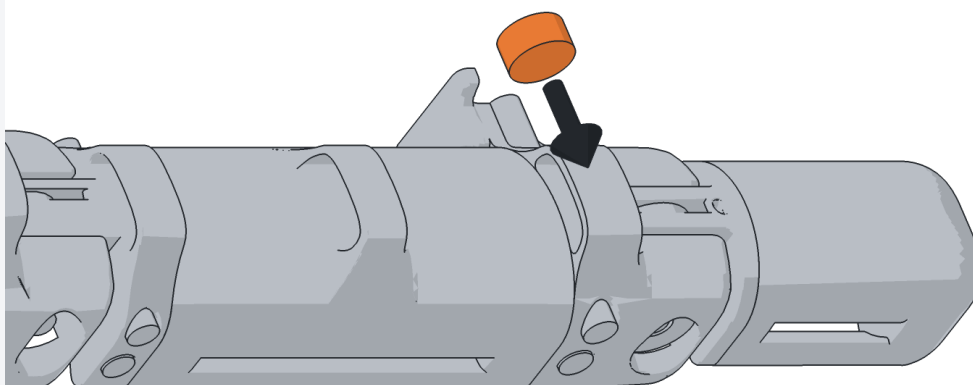
Fit the PIP centre piece



12

CLICKS PAST SPRINGS 1.25 MM³

Seat the PIP magnet

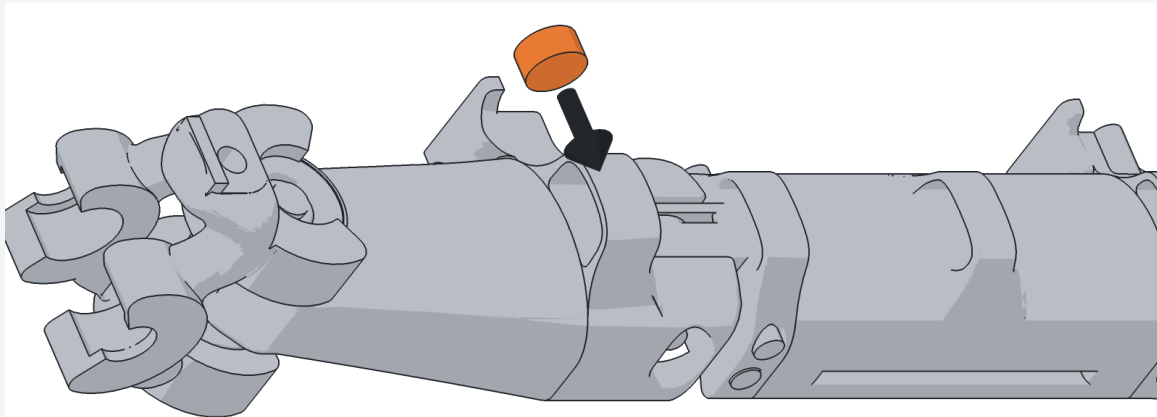


13

CLICKS PAST

SPRINGS 0.62 MM³

Seat the knuckle magnet

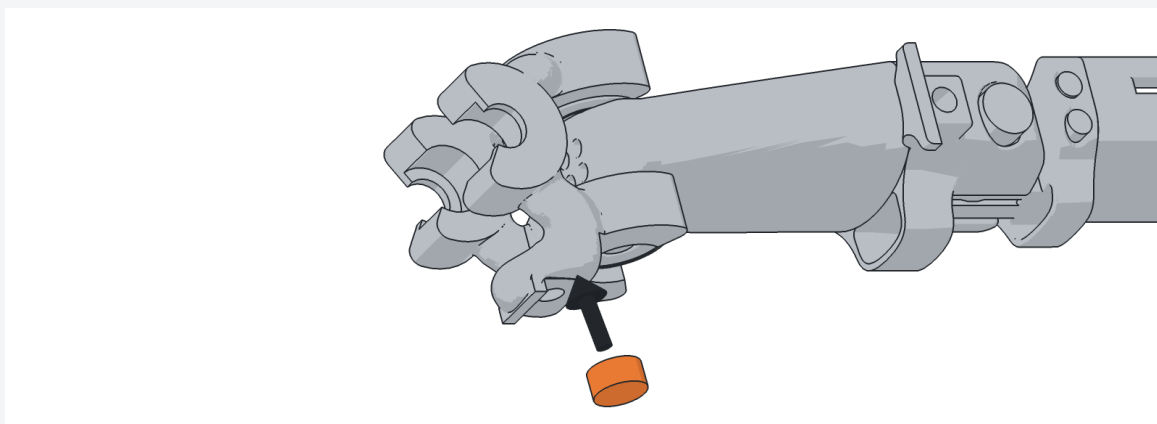


14

CLICKS PAST

SPRINGS 1.0 MM³

Set the third magnet on the shaft top

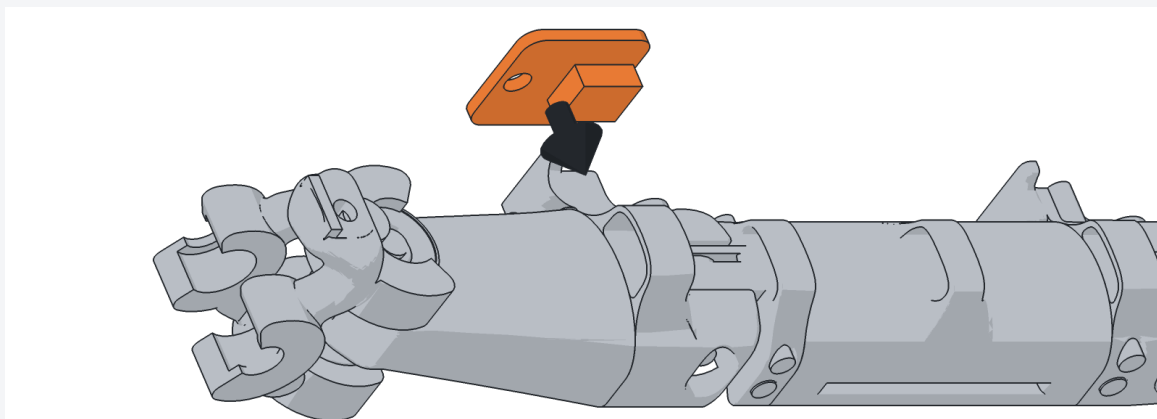


15

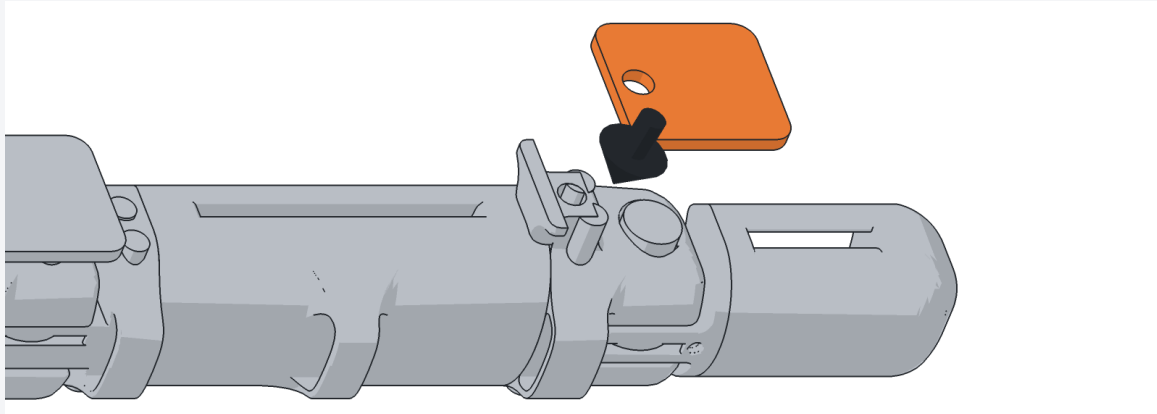
CLICKS PAST

SPRINGS 0.25 MM³

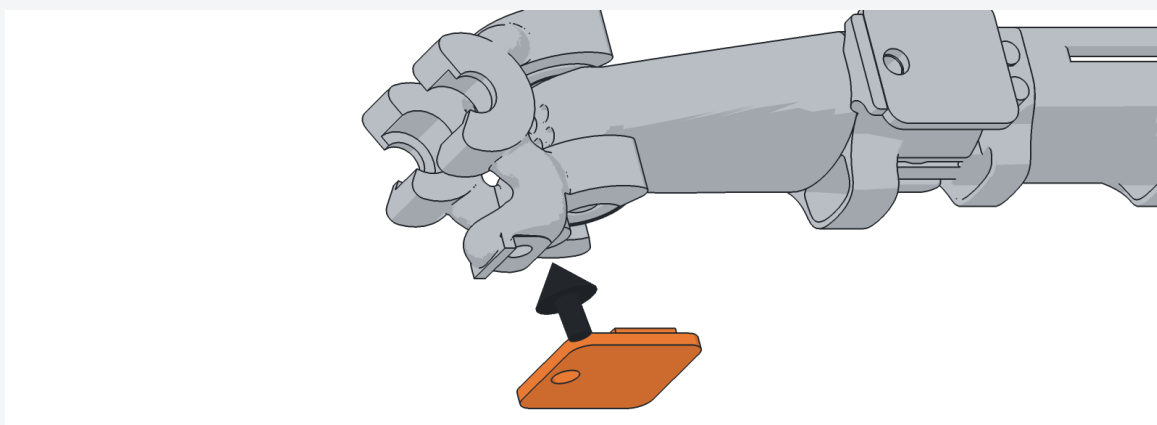
Fit the knuckle encoder board



16 Fit the MCP encoder board



17 Fit the PIP encoder board



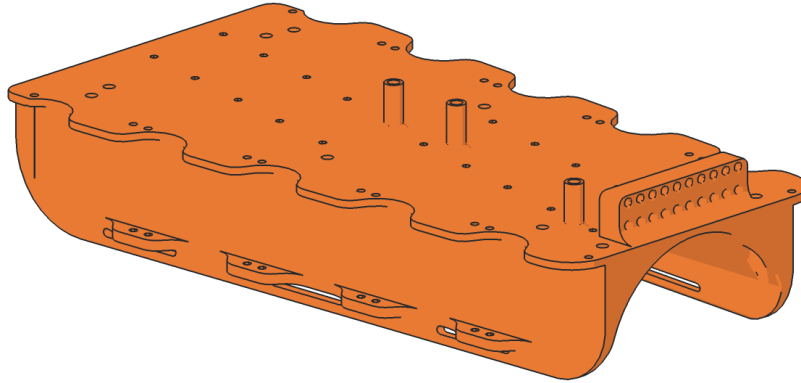
FOREARM HOUSING AND ELECTRONICS

The servos drop into their bays in any order. The deck carries the electronics, so it mounts right after them; the case closes over everything at the end.

1

START HERE

Start with the forearm housing



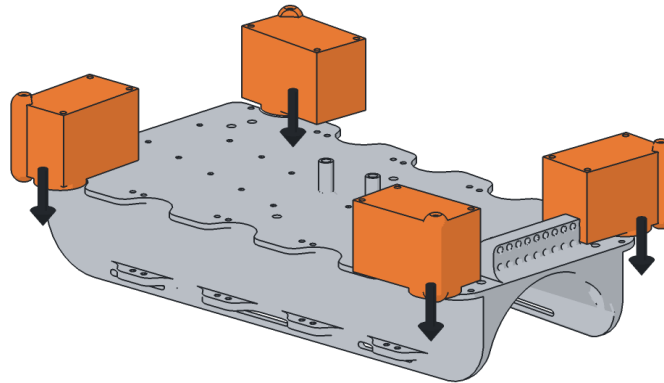
2

CLICKS PAST

4X M2 X 6 FROM BELOW

SPRINGS 1.88 MM³

Drop the corner pillars in

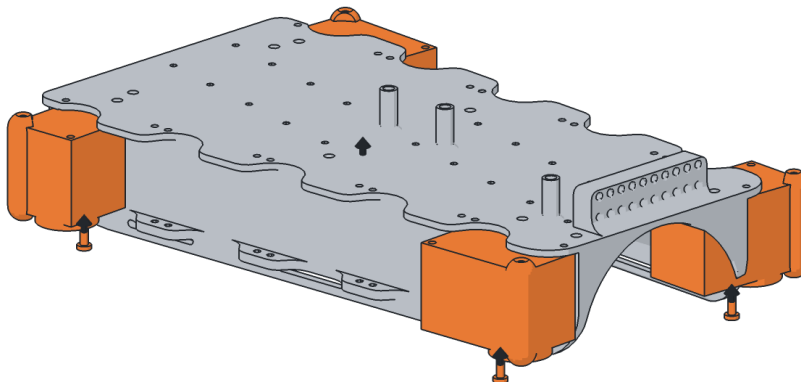


3

THEN THE SCREWS

4X M2 X 6 FROM BELOW

Screw it down



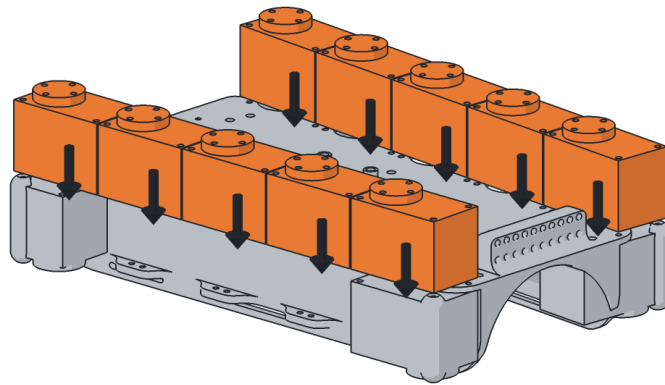
4

CLICKS PAST

10X XC330 - ANY ORDER

SPRINGS 8.12 MM³

Lower the ten servos into their bays



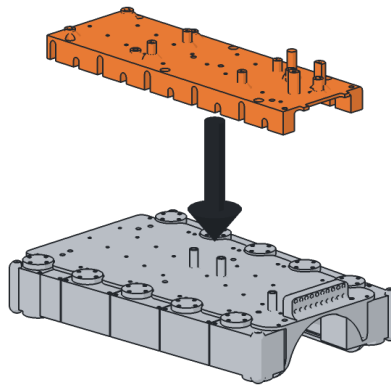
5

CLICKS PAST

6X M3 + 4X M2

SPRINGS 0.38 MM³

Mount the electronics deck

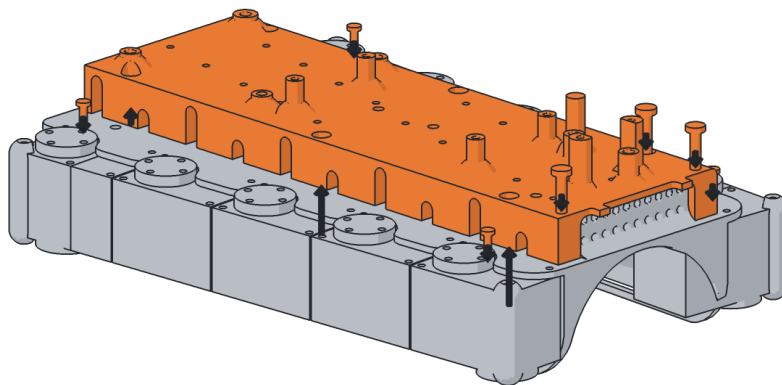


6

THEN THE SCREWS

6X M3 + 4X M2

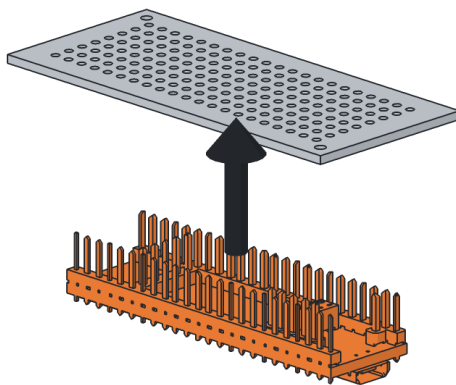
Screw it down



7

SOLDER

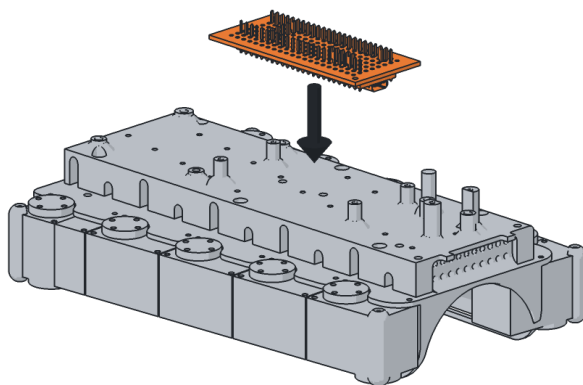
Solder the Teensy under its plate, pins up



8

M2 X 4 + M2 X 6 FROM ABOVE

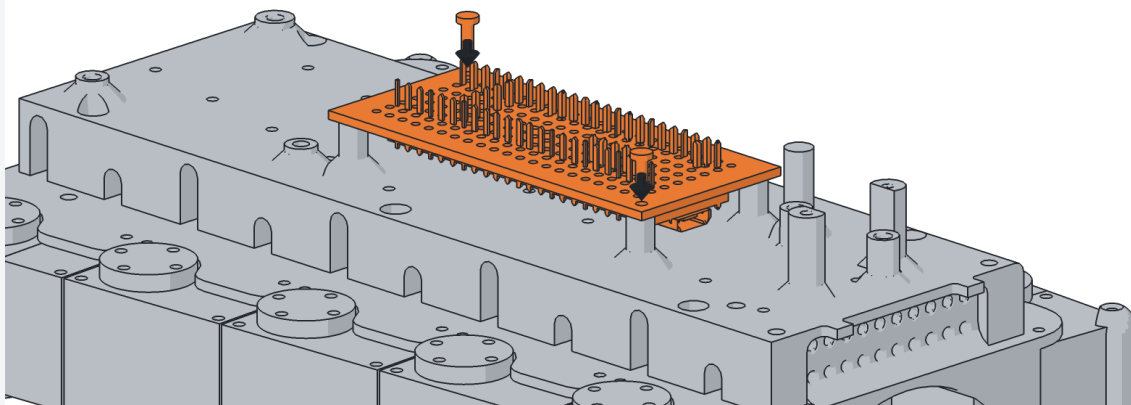
Install the Teensy plate



9

THEN THE SCREWS**M2 X 4 + M2 X 6 FROM ABOVE**

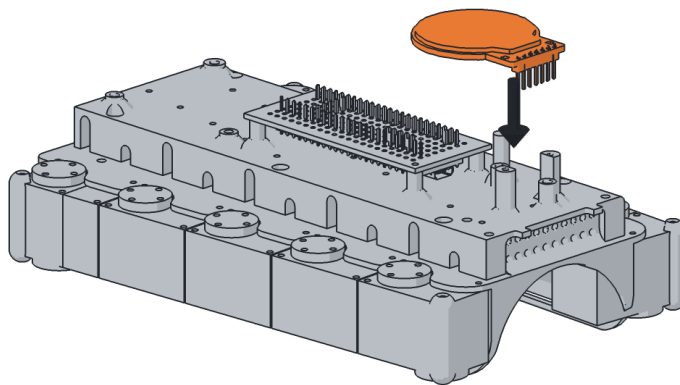
Screw it down



10

2X M2 X 6 FROM ABOVE

Fit the round display

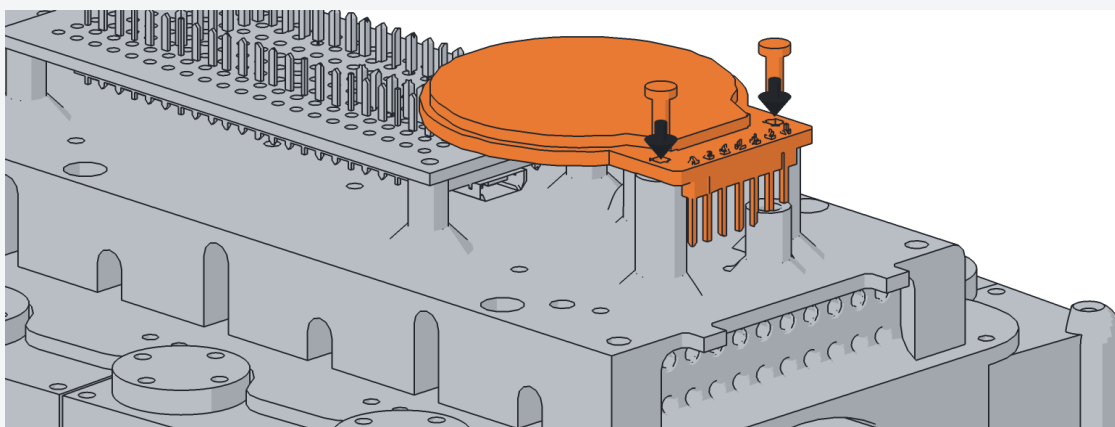


11

THEN THE SCREWS

2X M2 X 6 FROM ABOVE

Screw it down



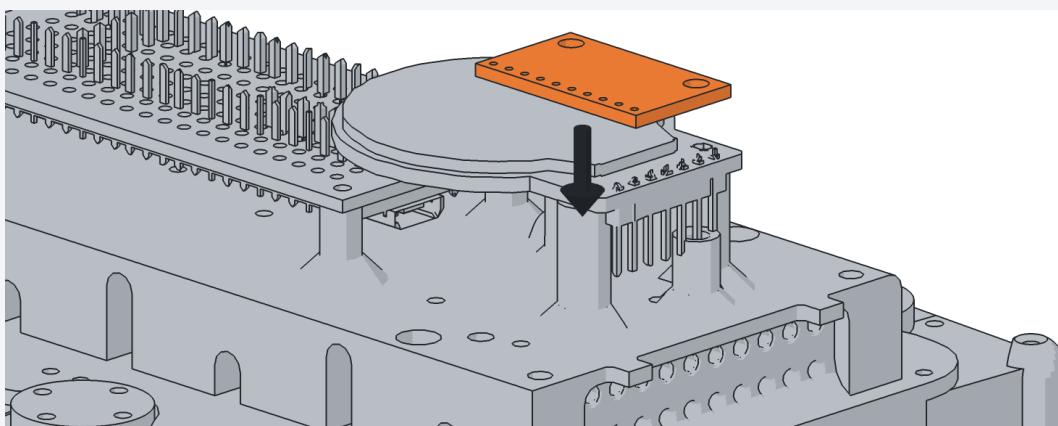
12

CLICKS PAST

2X M3 X 4 FROM ABOVE

SPRINGS 1.38 MM³

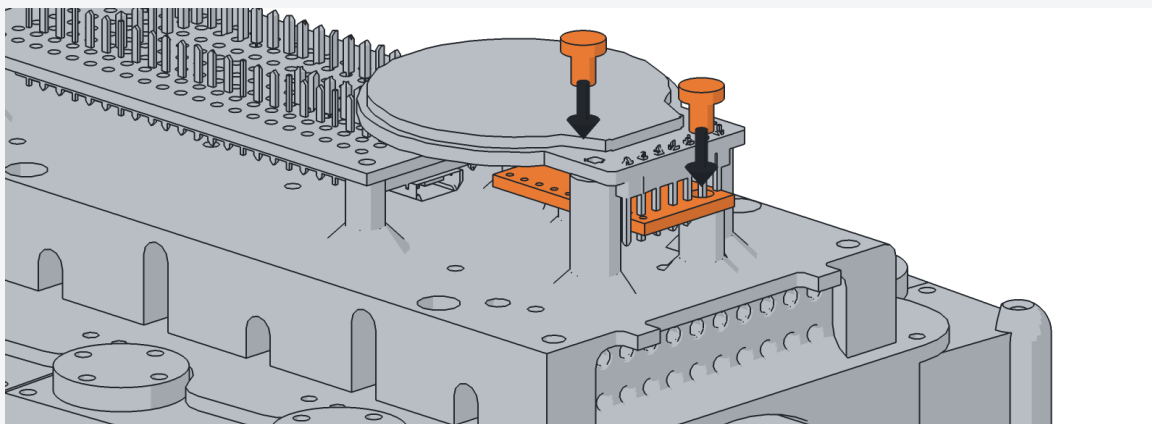
Fit the forearm IMU



13

THEN THE SCREWS 2X M3 X 4 FROM ABOVE

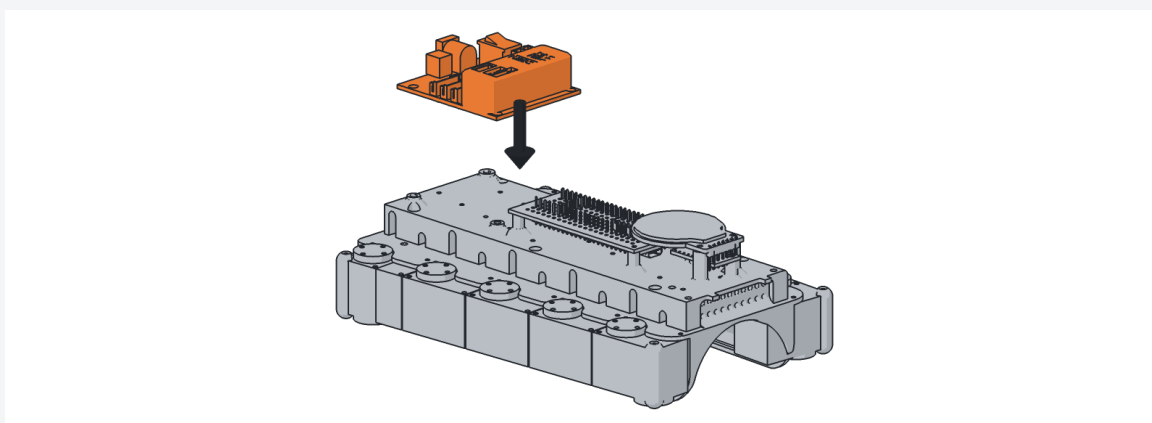
Screw it down



14

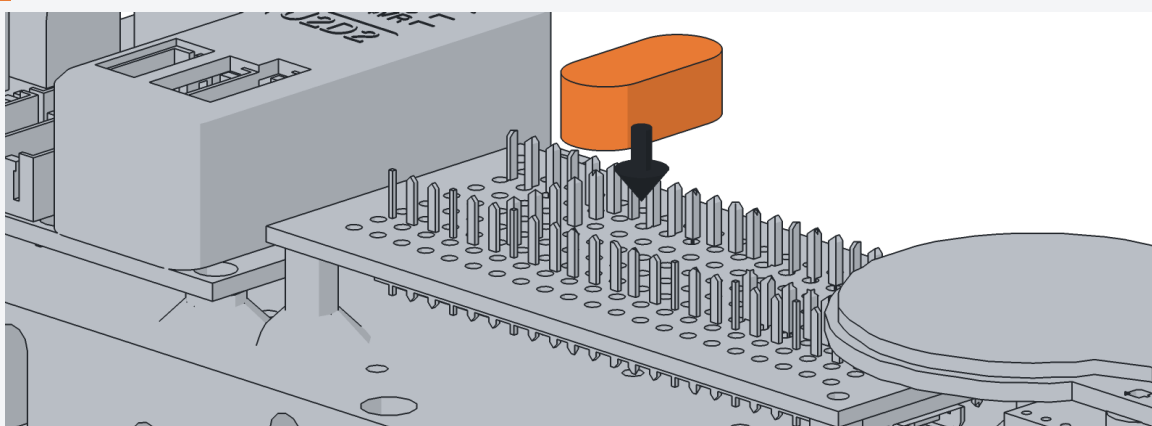
CLIPS IN - CONFIRM

Seat the power hub board

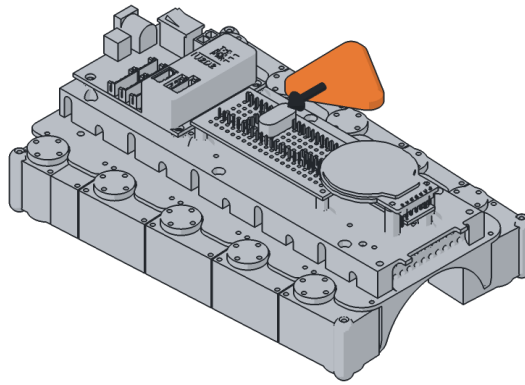


15

Fit the EMG holder



16 Fit the EMG module



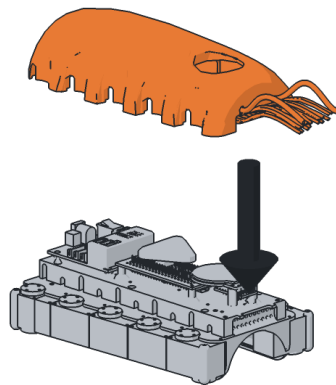
17

CLICKS PAST

FASTENERS - CONFIRM

SPRINGS 6.88 MM³

Lower the case over the device



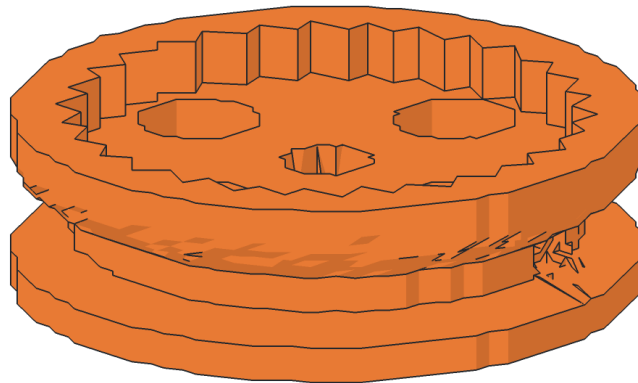
A TENDON SPOOL BAY

One bay of ten; five along each side. The ratchet sits between the spool and its cover.

1

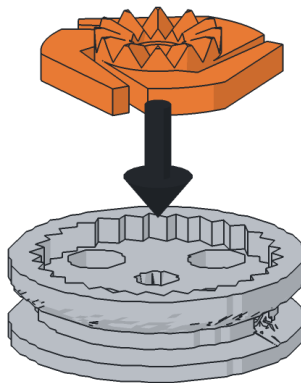
START HERE

Seat the spool on its servo



2

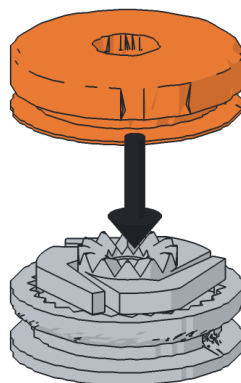
Set the ratchet on the spool



3

CLICKS PAST SPRINGS 10.38 MM³

Close the cover over the ratchet



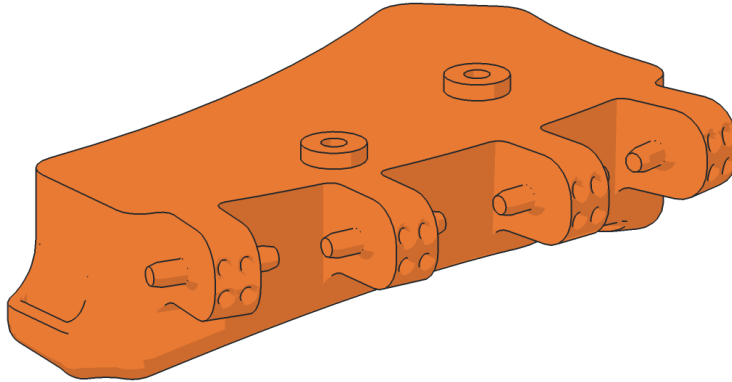
PALM STACK

The palm, the carrier board from the fab, two M2 spacers, the IMU soldered on, the cap.

1

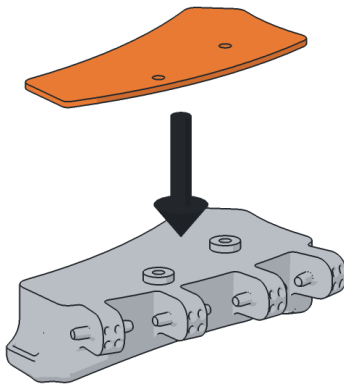
START HERE

Start with the palm



2

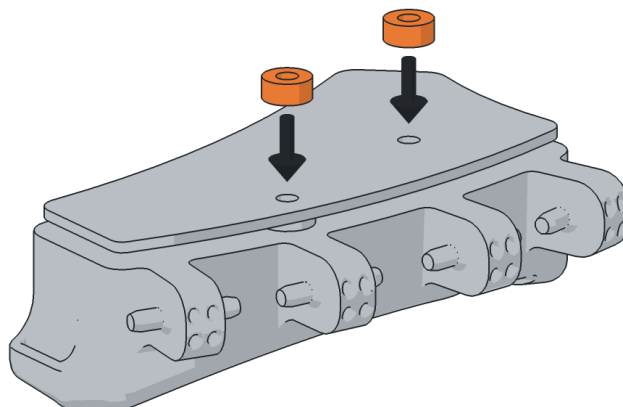
Fit the carrier board



3

2X M2 SPACER

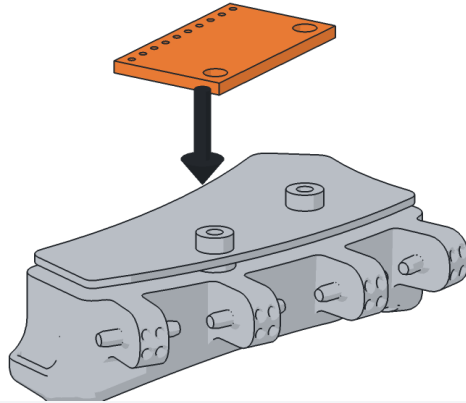
A spacer on each mount



4

SOLDER 2X M2 X 6

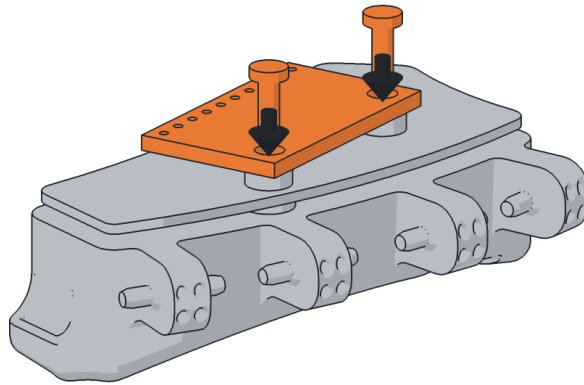
Solder the IMU onto the carrier



5

THEN THE SCREWS 2X M2 X 6

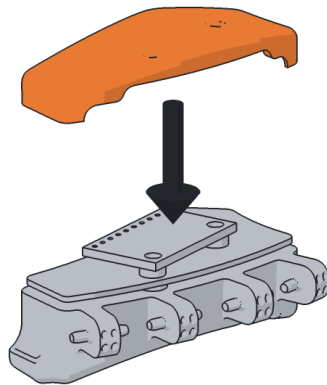
Screw it down



6

CLICKS PAST M2 X 8 + M2.5 X 8 **SPRINGS 2.12 MM³**

Close the palm cap

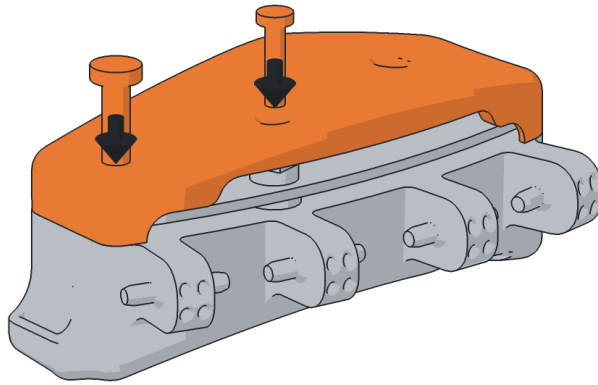


7

THEN THE SCREWS

M2 X 8 + M2.5 X 8

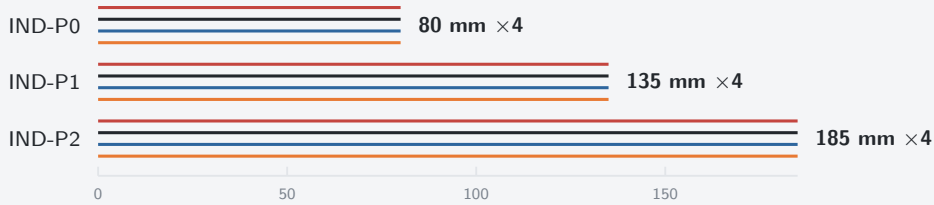
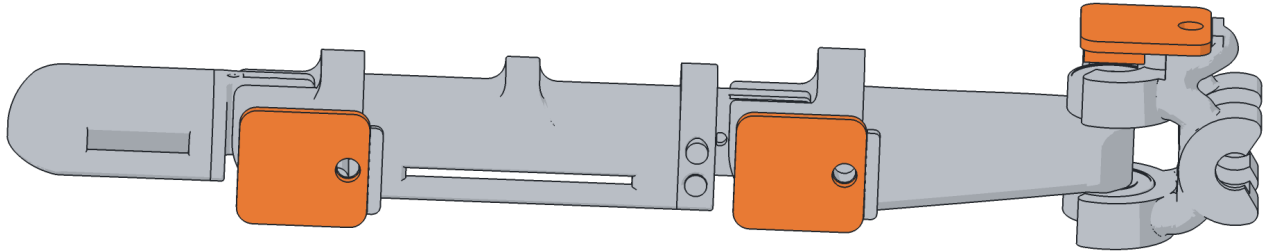
Screw it down



CUT THE FINGER WIRING

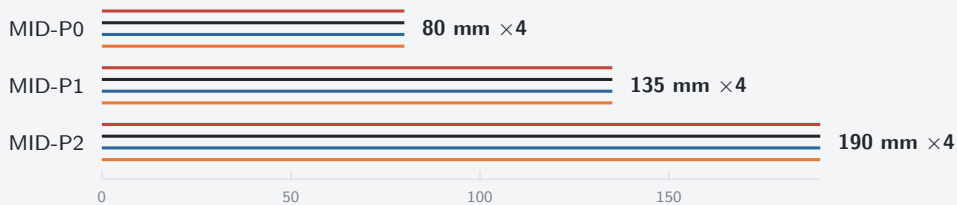
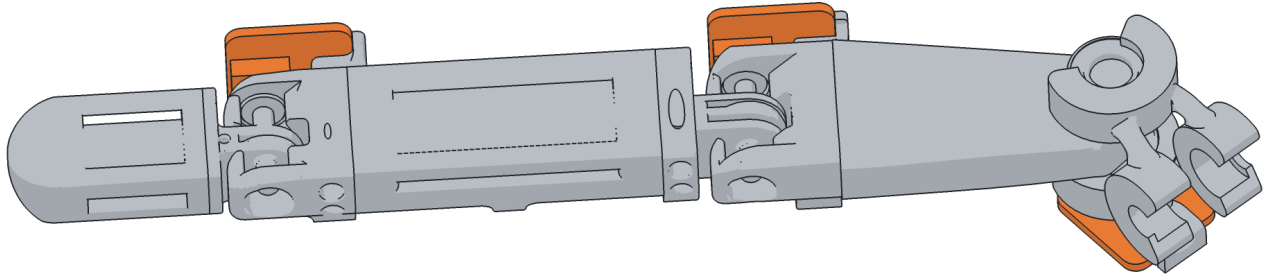
Every conductor is 30 AWG silicone-insulated stranded wire - never solid core. Four conductors per encoder board, both ends bare 1×2 mm solder pads, no connectors anywhere. Cut all twelve runs for a finger from its table; label each with the run name before you cut the next.

Index finger



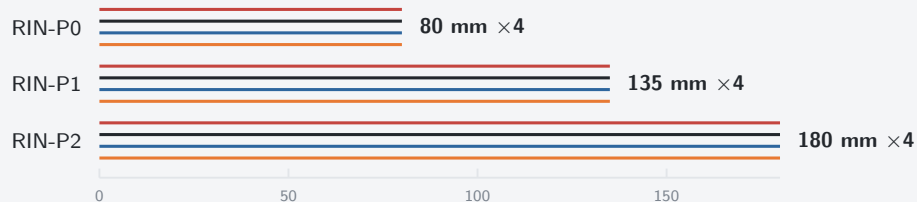
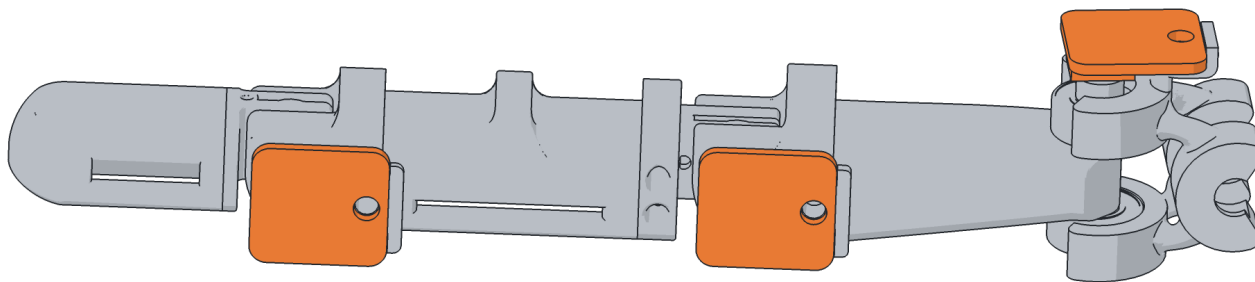
bars at 1:2 scale, ticks every 50 mm; whole finger: 1.60 m of wire.

Middle finger



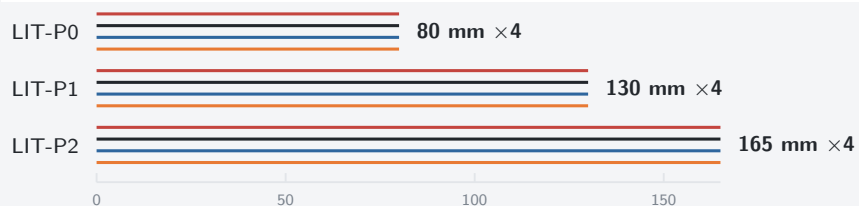
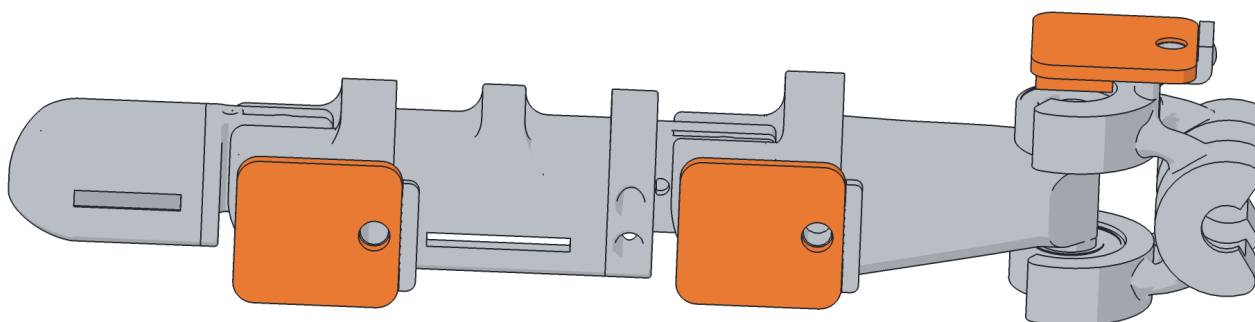
bars at 1:2 scale, ticks every 50 mm; whole finger: 1.62 m of wire.

Ring finger



bars at 1:2 scale, ticks every 50 mm; whole finger: 1.58 m of wire.

Little finger



bars at 1:2 scale, ticks every 50 mm; whole finger: 1.50 m of wire.

The wire itself, every run the same:



30 AWG stranded, silicone - shown 4:1

0.8 mm over insulation, 0.25 mm conductor; at 1:1:

Trunk, palm carrier to Teensy: **240 mm × 4** (SDA_Main, SCL_Main, 3V3, GND), the r=20 mm row of WIRE-SPEC §5. Colours: red 3V3, black GND, blue SDA, yellow SCL.

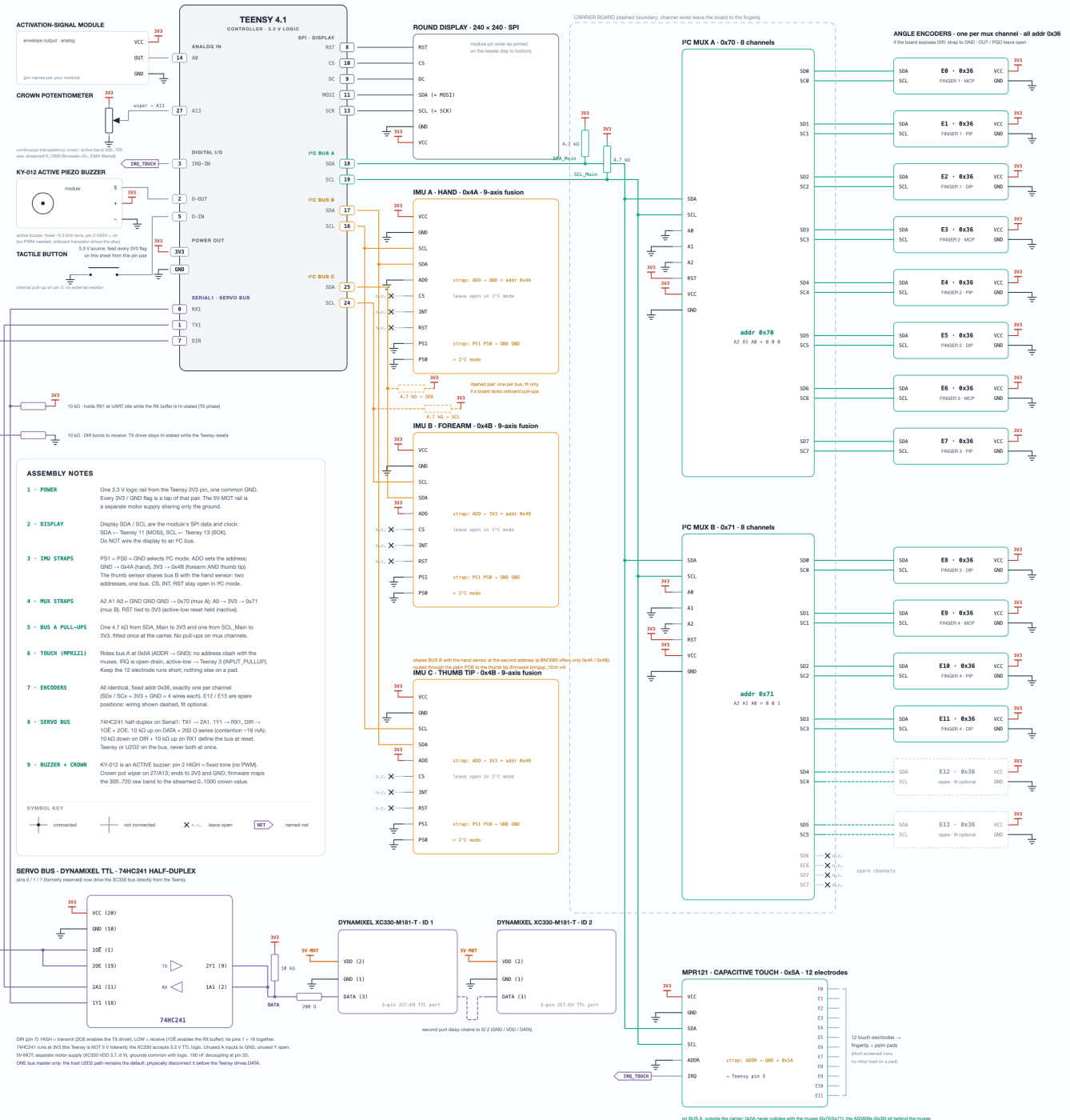
SOLDER THE BOARDS

The full schematic below is the drafted rev-2 sheet from the thesis work (2026-07-19). The table maps every finger run to the carrier pads it solders to - note the ring finger's third encoder crosses to mux B.

TAKTO ONE - GLOBAL WIRING SCHEMATIC

point-to-point, every pin terminated - KiCad conventions (junction dot = connect, crossing without dot = no connection, x = leave open)
rev 2: adds the thumb-tip IMU (bus B), MPR121 touch (bus A), crown potentiometer, KY-012 buzzer and the 74HC241 servo bus

PC bus A (encoders + touch) SPI (display) Dynamixel TTL (half-duplex) 3V3 net
PC bus B / C (IMUs) analog / digital 5 V motor rail GND net



Global sheet, rev 2 (supersedes wiring_schematic.png - pins, buses and addresses per DATA.md + bringup_15ch.v4 - 74HC241 circuit per the standard TTL half-duplex reference - verify strap conventions and module enclosures before first power-up - motors default to the host U202 path (single-master rule)

Source: Fable/wiring/global_wiring_schematic rev 2, 2026-07-19, drawn for this project; carrier nets per Zero_Palm_V2.kicad_sch.

WHERE EACH RUN LANDS

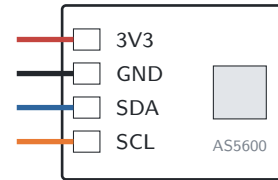
Every encoder is the same board at address 0x36; the mux channel is what tells them apart. Solder the run's SDA (blue) and SCL (yellow) to the listed channel pads on the carrier, 3V3 (red) and GND (black) to the rail pads beside them. Both ends are bare 1×2 mm pads - no connectors.

run	encoder	mux	carrier pads
IND-P0	E0	A · 0x70	SD0/SC0
IND-P1	E1	A · 0x70	SD1/SC1
IND-P2	E2	A · 0x70	SD2/SC2
MID-P0	E3	A · 0x70	SD3/SC3
MID-P1	E4	A · 0x70	SD4/SC4
MID-P2	E5	A · 0x70	SD5/SC5
RIN-P0	E6	A · 0x70	SD6/SC6
RIN-P1	E7	A · 0x70	SD7/SC7
RIN-P2	E8	B · 0x71	SD0/SC0
LIT-P0	E9	B · 0x71	SD1/SC1
LIT-P1	E10	B · 0x71	SD2/SC2
LIT-P2	E11	B · 0x71	SD3/SC3

← crosses to B

E12/E13 = spare channels (mux B SD4/SC4, SD5/SC5), left open

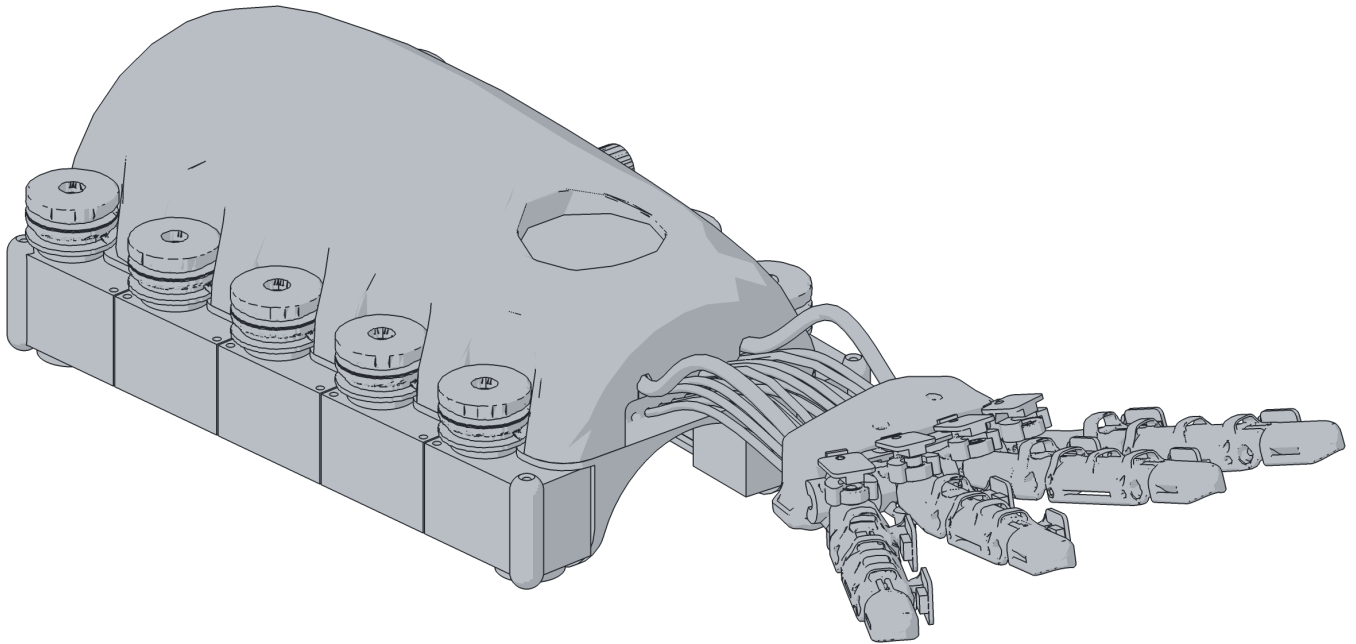
Encoder board, both ends alike



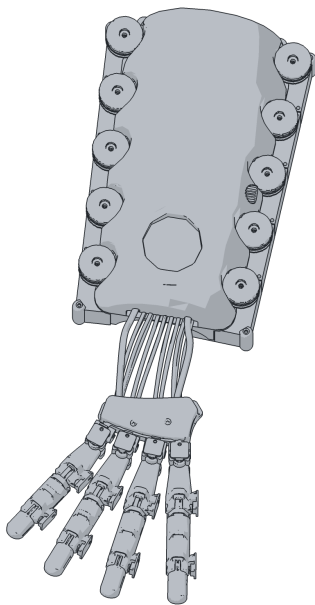
The palm IMU (BNO085, 0x4A) is a **soldered** joint onto the carrier (correction A13), on I²C bus B - not behind the muxes. Then its two M2×6 screws through the spacers.

THE FINISHED DEVICE

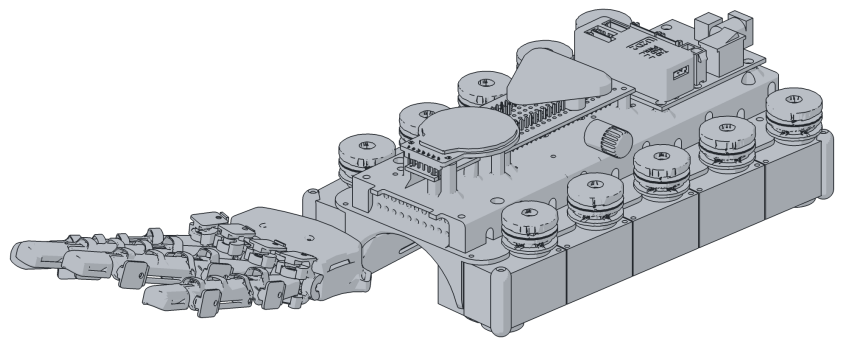
All four fingers built and instrumented, the case on. Every view on this page is rendered from the same CAD you just assembled.



Three-quarter view, case on.



From above: the four digits and their spread.



Case off: the electronics deck and the spool bays.

4 long fingers built and instrumented · **12** magnetic encoders · **10** tendon-spool bays · **2** XC330 servos fitted · **100 %** PETG as built

The housing is drilled and wired for eight servos across the four fingers, with two bays held for a future thumb. Two are fitted here: that is a budget limit, not a design one, and the spool bays, wiring and firmware already scale to the full count. The motor bus is owned by the host PC through a U2D2; the Teensy senses only.

CHECK YOUR PARTS AGAINST THESE

Print this page at 100 % and lay each real part on its outline. The grey bar at the bottom must measure exactly 50 mm; if it does not, your printer scaled the page.

AS5600 board



10.5 × 11.5 mm · 4 pads

MCP bearing MR84



Ø8 / bore Ø4 × 3 mm

PIP bearing



Ø5 / bore Ø2 × 2.5 mm

Magnet



Ø4 × 2 mm, diametric

M2 spacer



Ø4.5 / Ø2 × 2 mm

Guide rod, MCP



Ø2 × 13.2 mm · 2x

Guide rod, PIP



Ø2 × 16 mm · 2x

M2 x 4



1x · Teensy plate

M2 x 6



13x · everywhere

M2 x 8



1x · palm cap

M3 x 4



2x · forearm IMU

M2.5 x 8



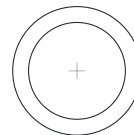
1x · palm cap

M3 x 16



6x · electronics deck

Tendon spool



Ø17 outline · 10 bays

Wire, 30 AWG



Ø1.2 insulated



exactly 50 mm at 100 %